

SKY130 • RV32I • 121 MHz

TRACE-V

A PicoRV32 RISC-V core on SKY130, built by agentic AI and verified with independent open-source tools

Traceable RISC-V: AI-built, Checked, Evidenced

8.25 ns clock 3 corners: ss / tt / ff 75 frozen evidence files

License

NON-COMMERCIAL ACADEMIC INSTRUCTION AND RESEARCH USE

The released PicoRV32 sky130 design files and results are licensed for non-commercial academic instruction and research. Prior written permission is required for any other use.

Third-party components keep their own licenses

picorv32.v is licensed under ISC, © 2015 Claire Xenia Wolf.
The SkyWater SKY130 PDK and standard-cell data are licensed under Apache-2.0 by SkyWater Technology Foundry and Google LLC.

Origin

The outputs were produced with EDA software licensed to Purdue University for academic instruction and research, and the license keeps the outputs within that use.

Research collaboration or licensing inquiries about the framework are directed to Raisul Islam — raisul@purdue.edu.

Background: PicoRV32

THE CORE

- PicoRV32 is a size-optimized RISC-V CPU core by Claire Xenia Wolf, released under the ISC license [1].
- The core can be configured as RV32E, RV32I, RV32IC, RV32IM or RV32IMC [1].
- On Xilinx 7-series devices, the core runs at 250–450 MHz in 750–2,000 LUTs, with a CPI of about 4 [1].
- PicoRV32 is a general-purpose CPU, and RV32I is the 40-instruction RISC-V base integer ISA [2].

THIS BUILD: DEFAULT PARAMETERS

- The default parameters are used: RV32I, with 32 registers and counters [1].
- Multiply/divide, compressed instructions and interrupts are not included [1].
- Misaligned accesses and illegal instructions cause a trap [1].

Process: the SkyWater SKY130 open 130 nm PDK is used, with sky130_fd_sc_hd cells [3].

The design

PicoRV32

The core is a 32-bit RISC-V (RV32I) design by Claire Xenia Wolf, under the ISC license.

SkyWater sky130A

The layout holds 87,237 transistors in 7,845 standard cells of the open sky130_fd_sc_hd library.

8.25 ns · 121 MHz

The clock uncertainty is 0.25 ns at every corner.

ss · 100 °C · 1.60 V

tt · 25 °C · 1.80 V

ff · -40 °C · 1.95 V

COMMERCIAL IMPLEMENTATION



Open checkers are used for physical signoff, so that anyone can re-verify the results: magic DRC, KLayout DRC (sky130A_mr deck), magic antenna, LVS with magic and netgen, and gate-level simulation with Icarus Verilog on the post-layout SDF.

Routed layout

357.42 × 355.64 μm

die

7,845 cells

176 are clock-tree buffers; 8,451 nets

71,909 μm²

cell area

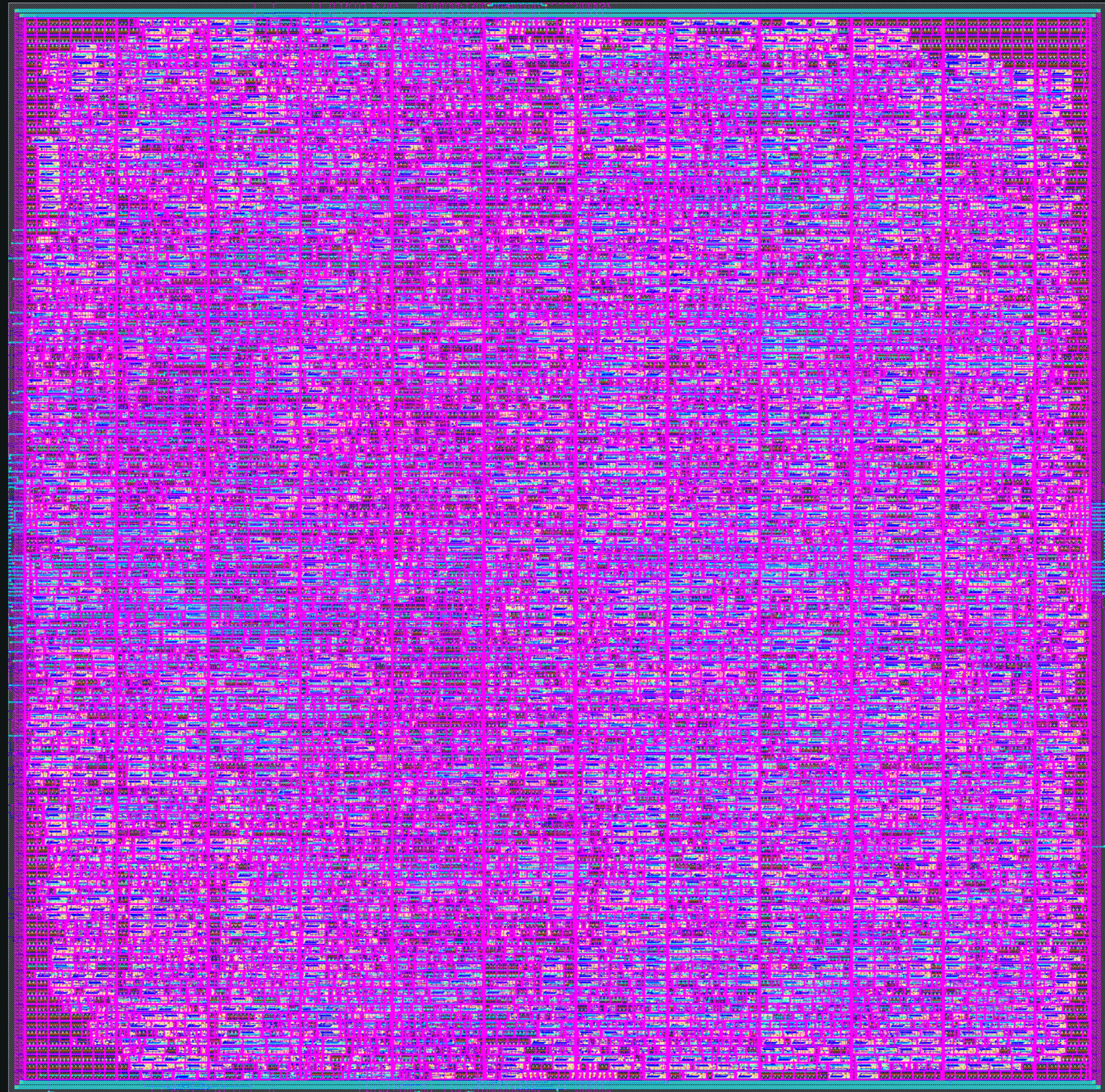
0.86–0.96 ns

clock-tree insertion delay

13.0 mW

static, vectorless, 20% input activity at 8.25 ns

05 · KLayout render of the final GDS



Signoff at 8.25 ns: physical and logical

Check	Result
DRC, magic	0 violations
DRC, KLayout (sky130A_mr deck)	0 findings
Process antenna, magic	0 violations
LVS, magic + netgen, against the routed netlist	matched; 8,492 / 8,492 nets
LEC, RTL → synthesized netlist (Conformal)	equivalent; 1,916 compare points, 0 non-equivalent, aborted or unknown
LEC, synthesized → routed netlist (Conformal)	equivalent; 1,864 compare points, 0 non-equivalent, aborted or unknown

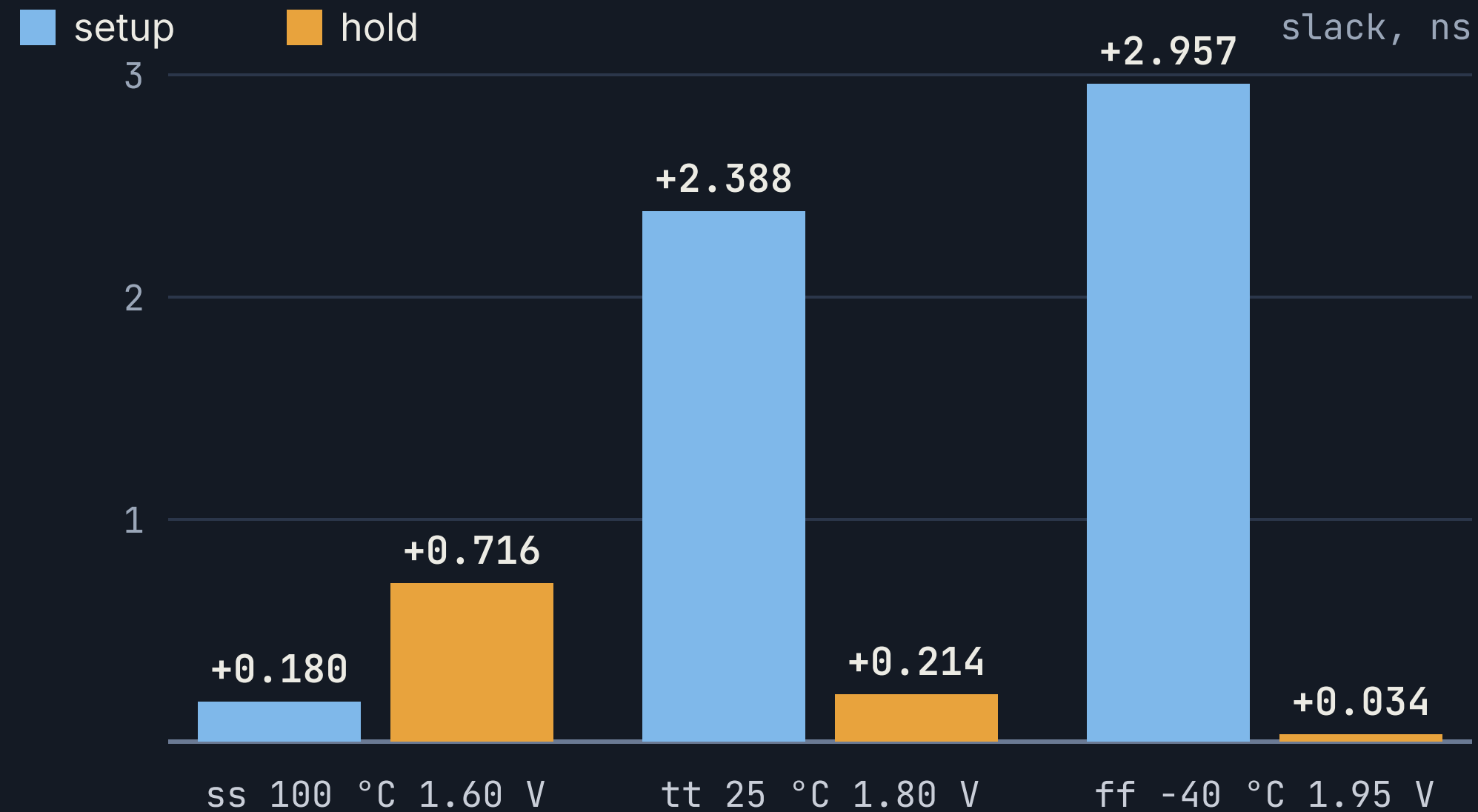
LVS is run against the routed netlist and LEC is run twice, so that the output of each tool is also checked independently.

Signoff at 8.25 ns: timing, simulation, power

Check	Result
Setup slack, ss / tt / ff	+0.180 / +2.388 / +2.957 ns; 0 violating endpoints
Hold slack, ss / tt / ff	+0.716 / +0.214 / +0.034 ns; 0 of 1,626 endpoints violating, input ports included
Max transition / capacitance / fanout	0 / 0 / 0 at all three corners
Golden program on RTL	pass
Golden program, post-layout gate level, ss SDF, 8.25 ns	pass; the 4.125 ns control fails
Gate-level sim with timing checks (Xcelium), 8.25 ns	pass, 0 violations; 3,114 setup/hold checks applied (100%); 4 ns control fails with 76 timing violations
Post-layout timing wall	passes at 8 ns, fails at 6 ns

Setup and hold include the 0.25 ns uncertainty. The golden program is hand-assembled RV32I: it sums 1 to 10 and stores 55. Static IR is shown on slide 16.

Slack per corner



0 violating

No setup endpoint violates, and 0 of 1,626 hold endpoints violate, input ports included.

Setup and hold slack both include the 0.25 ns uncertainty.

How the platform works

01

Requirements and implementation

The requirements were set once by the designer; synthesis, place and route, signoff and debugging were carried out by agentic AI.

02

Traceable figures

Every figure is traceable to the run and input files that produced it.

03

75 frozen evidence files

Cited evidence is frozen and checksummed.

04

Every physical, DRV and simulation check has a control

Every physical, DRV and simulation check was shown to fail on a known-bad input before it was trusted.

05

No silent passes

A check that cannot run is recorded as a failure.

06

Independent checkers

Layouts built by Innovus are checked by magic, KLayout and netgen; LEC links the routed netlist back to the RTL.

DEFECTS FOUND DURING SIGNOFF

Six defects that passed the tools' own reports, and how each was found

Each result was read back from the design and compared with the requirement.

Results that should agree were compared.

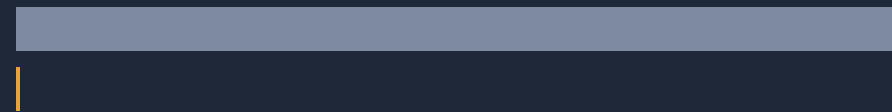
Each check was shown to fail on a known-bad input before it was trusted.

Defect 1: the fanout limit was ignored

A maximum fanout of 10 was set in the SDC. Innovus finished with a 240-pin fanout violation, and the flow completed normally.

fanout violation

240 → 0



clock-tree buffers

23 → 174



insertion delay (CTS average)

0.46 → 0.95 ns



Gray bar: before the fix. Amber bar: after.

HOW IT WAS FOUND

The fanout count was read back from the routed design and compared with the limit declared in the SDC. Signoff was refused at 240 pins over, although Innovus reported success.

The two settings that Innovus needs (optimizer and clock-tree fanout repair) were found by agentic AI, the route was re-run, and the count is now 0.

Defect 2: STA passed, simulation trapped

mem_ready → reg_op1

-1.87 ns

The interface was timed as the testbench drives it: memory is driven on the falling edge, which leaves half a cycle.

1. After the fanout fix, setup was met by STA at 8 ns, but the post-layout simulation trapped at 8 ns.
2. An instruction that had been given correctly was misread by the core.
3. The cause was the absence of input and output delays in the SDC, so the memory interface was given a full cycle by STA instead of the testbench's half cycle.

HOW IT WAS FOUND

Two results contradicted each other: STA met, simulation failed. The STA pass was not accepted. The wrong transaction was traced by agentic AI, the interface was timed the way the testbench drives it, and the constraint was fixed instead of the test.

After the re-run, the simulation passes at 8 ns and the input-port hold violations (down to -0.92 ns before) are gone.

Defects 3 and 4: faults in the checks themselves

DEFECT 3

The gate-level simulation was all-X

The supplies of the power-aware cell models were unconnected, so every simulation failed, at every clock.

CONTROL

The defect was found by a zero-delay positive control: the same netlist with no delays must pass.

DEFECT 4

The antenna check could not fail

The initial negative control contained a diode that was not needed, so the check passed it.

FIX

The control was replaced by a frozen known-bad layout that must fail.

HOW THEY WERE FOUND

Every check carries a control that it must pass or fail. When a control misbehaved, the check was marked unproven instead of clean, and the check itself was repaired by agentic AI before any result from it was trusted.

Defects 5 and 6: found in the 8.25 ns re-run

DEFECT 5

A cell excluded from synthesis was re-inserted by place and route.

The and2_0 cell was excluded in Genus, but Innovus restructuring re-inserted it. One instance abutted a neighbor and produced a KLayout npc.2 spacing violation (0.16 μm against 0.27 μm), while magic DRC was clean.

FOUND AND FIXED

The defect was found by the second DRC deck. It was fixed by applying the exclusion in Innovus as well, with a post-route count of excluded cells that must be zero.

DEFECT 6

An antenna violation remained after the router's repair.

One net was left at a ratio of 403.85 against a limit of 400.

FOUND AND FIXED

The defect was found by the open antenna check. It was fixed by repairing to 96% of the limit.

What it cost

1 Oct → 7 Oct 2026

The calendar time runs from the start of the commercial flow on the open sky130 PDK to block signoff (7 Oct, 16:15 EDT).

38 min · 45 min

One full signoff pass takes 38 min of compute and 45 min end to end.

62 jobs · 5.1 h

62 cluster jobs ran from 1 Oct through the 8 ns signoff, with 5.1 h of total wall time. The 8.25 ns re-run added 21 cluster jobs and 2.0 h of wall time.

2,907 calls

Agentic AI usage was 2,907 calls, 1.26 M output tokens, 0.89 B cached input tokens read, 32 M written to cache and about 6.5 k uncached.

About 3.1 hours of designer interaction were spent over 7 days, in 86 messages. The requirements were set and each decision was approved by the designer. Implementation, verification, debugging and fixes were carried out by agentic AI.

Times are measured from message timestamps, with gaps under 20 minutes counted as work. Token and time figures are upper bounds, because parallel work outside this design was included.

Static IR drop: 11 mV on VPWR

11 / 11.6 mV

maximum drop, VPWR / VGND

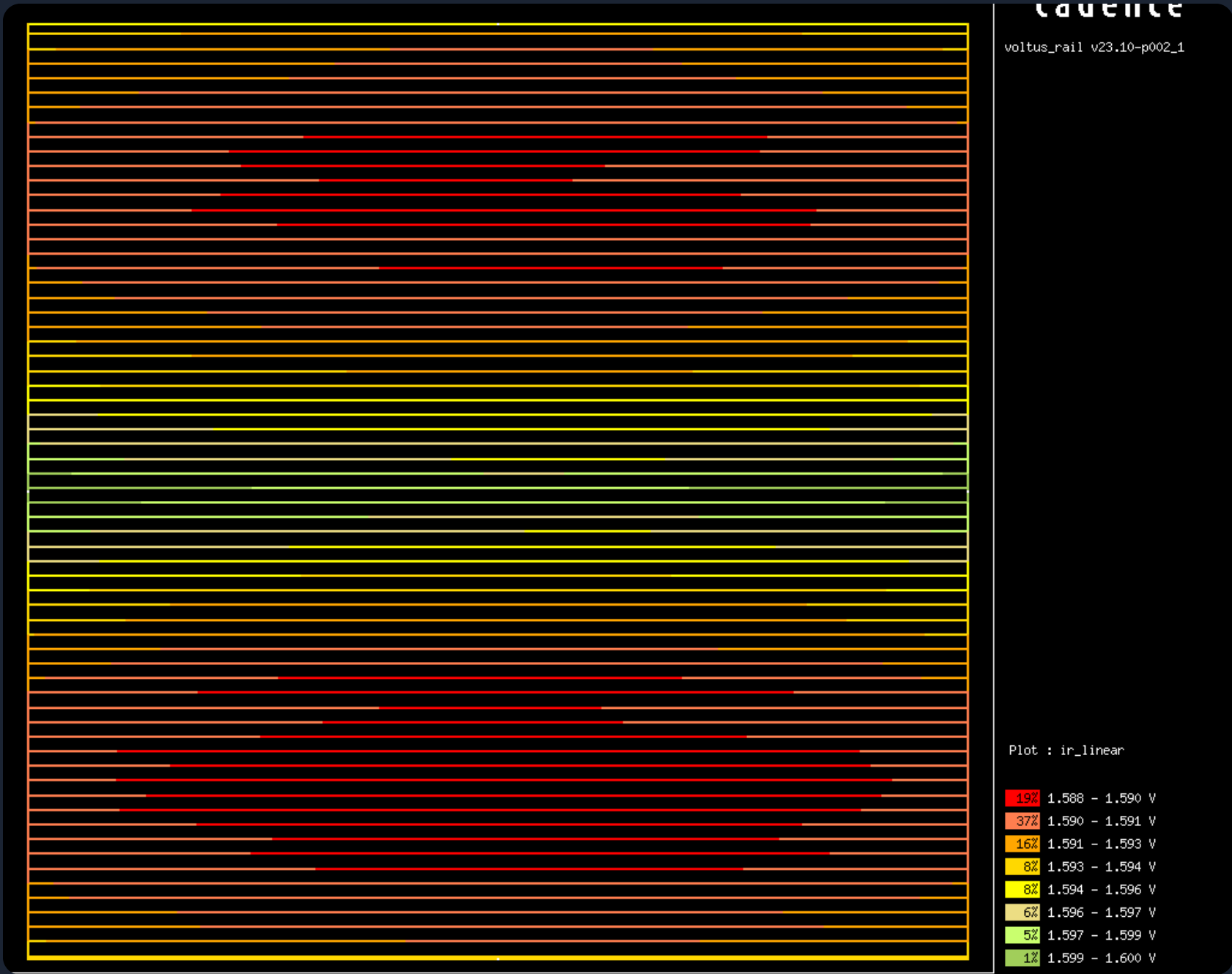
80 mV each

budget, 5% of 1.6 V

1.588–1.600 V

VPWR across the die, 8.25 ns route

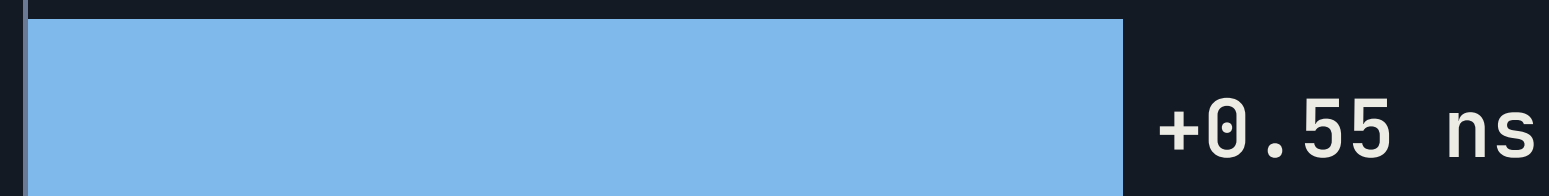
The analysis is static and vectorless, with 20% input activity at 8.25 ns.



Cross-check: RC corners

The cross-check was run with an independent extractor, OpenRCX, and with OpenSTA: ss setup at 8.25 ns.

Nominal RC



Max RC



Max RC costs 0.21 ns.

The ff hold with min RC is +0.03 ns, which is met.

Signoff extraction is at nominal RC, because the Cadence sky130 kit provides a typical extraction file.

This is an estimate combining two extractors: the 0.21 ns applied to the +0.180 ns signoff margin gives about -0.03 ns, so **about 8.3 ns** would be needed at max RC.

Limitations

1 Signoff is at nominal RC; the max-RC estimate is given on slide 17.

2 The I/O timing is set to the testbench's half-cycle memory protocol.

3 The margins are thin: ss setup +0.180 ns and ff hold +0.034 ns, after 0.25 ns uncertainty.

4 The post-layout simulation runs one 9-instruction program covering 6 of 40 RV32I opcodes; LEC and STA cover the whole design.

Scope: Packaging, electromigration, metal fill, pad frame, foundry precheck and dynamic IR were outside the scope of this block-level release.

Sources

[1] C. X. Wolf, PicoRV32, github.com/YosysHQ/picorv32.

[2] A. Waterman, K. Asanović (eds.), The RISC-V Instruction Set Manual, Vol. I, RISC-V International.

[3] SkyWater SKY130 PDK, github.com/google/skywater-pdk, built with open_pdks 1.0.471 (github.com/RTimothyEdwards/open_pdks).

Tools Cadence Genus, Innovus, Tempus, Conformal, Voltus, Xcelium; open checkers magic, KLayout, netgen, OpenSTA, OpenRCX, Icarus Verilog.

What this release covers

Timing was signed off at three corners with 0.25 ns clock uncertainty.

Every physical, DRV and simulation check was shown to fail on a known-bad input before it was trusted.

DRC was run with two independent decks, LVS against the routed netlist, and logic equivalence at both synthesis and routing.

Every reported figure is traceable to frozen, checksummed evidence.

Post-layout gate-level simulation was run with flop timing checks, and static IR was analyzed.

The physical, timing and simulation checks can be reproduced from the public package with open-source tools.